

ORIGINAL ARTICLE



Blood Pressure Change and Subclinical Target Organ Damage in Mid-Adulthood

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BACKGROUND: Blood pressure (BP) is a dynamic trait associated with cardiovascular disease. We aimed to estimate age-specific BP levels and rates of change from childhood to mid-adulthood and to examine their associations with subsequent subclinical target organ damage.

METHODS: We included 2508 participants from the Hanzhong Adolescent Hypertension Study with BP measured $\geq 4\times$ from 1987 to 2023. Surrogate markers of target organ damage were assessed, including arterial stiffness, left ventricular hypertrophy, and albuminuria. Growth models were used to construct BP trajectories and estimate age-specific BP levels and rates of change (slopes).

RESULTS: Rates of change in BP at each age point from childhood to mid-adulthood were positively associated with arterial stiffness and albuminuria in mid-adulthood, independent of corresponding BP levels. The magnitude of the associations rose from childhood, peaked in adolescence, and declined thereafter. For example, odds ratios (95% CIs) per 1 SD increase in systolic BP change rates for arterial stiffness increased from 1.94 (1.69–2.24) at age 6 to 2.11 (1.82–2.44) at age 13, then declined to 1.13 (1.01–1.28) by age 52. Faster BP increases were more strongly associated with arterial stiffness and albuminuria than concurrent BP levels during childhood and adolescence, whereas the opposite trend was observed in young and mid-adulthood. Similar age-dependent trends were identified for left ventricular hypertrophy, with minor variations in the ages at which associations reached statistical significance.

CONCLUSIONS: Accelerated BP increases during childhood and adolescence show a stronger association with mid-adult subclinical target organ damage than increases occurring in adulthood. (*Hypertension*. 2026;83:2224–2234. DOI: 10.1161/HYPERTENSIONAHA.125.26364.) • **Supplement Material.**

Key Words: adolescent ■ albuminuria ■ blood pressure ■ child ■ life cycle stages ■ middle aged ■ vascular stiffness

Cardiovascular disease (CVD) remains the leading cause of global morbidity and mortality, accounting for an estimated 20.5 million deaths and 422 million disability-adjusted life years in 2021.¹ Although substantial progress has been made in prevention and treatment, the global burden of CVD continues to increase. This trend underscores an urgent need to identify and mitigate risk factors much earlier in the life course. Blood pressure (BP), a key modifiable determinant of CVD,¹ exerts cumulative effects across the lifespan. Adverse

BP exposure may impair cardiovascular health decades before clinical disease manifests. Therefore, understanding how BP influences early subclinical alterations in cardiovascular structure and function is critical for developing effective life-course prevention strategies.

Hypertension-related target organ damage involves blood vessels (eg, arterial stiffness), the heart (eg, left ventricular hypertrophy [LVH]), and the kidneys (eg, albuminuria). Among these, arterial stiffness and LVH are well-established subclinical markers of CVD.^{2,3}

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/HYPERTENSIONAHA.125.26364>.

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Hypertension is available at www.ahajournals.org/journal/hyp

NOVELTY AND RELEVANCE

What Is New?

Although blood pressure (BP) is a dynamic trait and a key determinant of cardiovascular disease, it remains unclear whether specific crucial periods exist during the life course when rapid BP changes exert a disproportionately stronger influence on future target organ damage.

What Is Relevant?

The rates of change in BP at each age from childhood to adulthood were positively associated with subclinical target organ damage in mid-adulthood, independent of concurrent BP levels.

These associations were more pronounced during childhood and adolescence than in adulthood. During childhood and adolescence, the impact of BP change rates on later outcomes was stronger than that of absolute BP levels at the same age.

Clinical/Pathophysiological Implications?

The dynamic pattern of BP changes throughout childhood and adolescence, rather than a single BP measurement, may be crucial for identifying individuals at increased risk for subclinical cardiovascular disease later in life. Childhood and adolescence may represent a crucial window for BP-focused prevention and control strategies aimed at reducing future cardiovascular risk.

Nonstandard Abbreviations and Acronyms

AUC	area under the curve
baPWV	brachial-ankle pulse wave velocity
BP	blood pressure
CVD	cardiovascular disease
DBP	diastolic blood pressure
LVH	left ventricular hypertrophy
LVM	left ventricular mass
LVMi	left ventricular mass index
OR	odds ratio
SBP	systolic blood pressure

Albuminuria, an early sign of chronic kidney disease, is now also reckoned as a major risk factor for cardiovascular events.⁴ Elevated urinary albumin-to-creatinine ratio reflects widespread vascular dysfunction and has been integrated into the 2023 American Heart Association's Predicting Risk of CVD EVENTS calculator to improve risk assessment.⁵ Several longitudinal studies, including ours, have demonstrated that high BP levels in childhood and certain BP trajectory patterns from childhood to adulthood are associated with an increased likelihood of developing later-life arterial stiffness,^{6–8} LVH,⁸ and albuminuria.⁹ Nevertheless, little is known regarding whether there are crucial periods across the lifespan during which dynamic changes in BP confer lasting and disproportionate adverse effects on target organs.

While BP levels generally increase with age, the instantaneous rates of change at each age point vary among the population, and steeper increases may potentially lead to higher risks later in life.¹⁰ Some prior studies have explored the associations between the rates of change in BP during

specific life stages (either childhood or adulthood) and subsequent LVH,^{11,12} but none have compared these associations across different periods spanning from childhood to adulthood or evaluated their effects on arterial stiffness and albuminuria. Clarifying this may provide valuable insights into the optimal timing for early screening and the timely implementation of targeted prevention.

Using data from the ongoing Hanzhong Adolescent Hypertension Study cohort, we aimed to estimate BP levels and rates of change at each age from childhood through mid-adulthood and to examine their associations with subclinical target organ damage (including arterial stiffness, LVH, and albuminuria) in mid-adulthood. We hypothesized that higher age-specific BP levels and a faster rate of BP increase across the life course would be associated with increased risks of subclinical vascular, cardiac, and renal damage in mid-adulthood, and that the strength of these associations would differ across specific developmental periods.

METHODS

Data Availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request. A [Major Resources Table](#) is provided in the [Supplemental Material](#).

Study Cohort

The Hanzhong Adolescent Hypertension Study is an ongoing community-based cohort of 4623 school-aged children and adolescents recruited in 1987 from 26 survey sites across 3 towns (Qili, Laojun, and Shayan) in Hanzhong City, Shaanxi Province, China. At baseline, participants were selected based on the following criteria: elementary or middle school students, no history of chronic diseases, ability to communicate

effectively in Mandarin, and voluntary participation. Follow-up visits occurred in 1989, 1992, 1995, 2005, 2013, 2017, and 2023. In 2005, a subsample of 436 participants was selected using systematic sampling: all participants were ordered by unique identification number, and every tenth individual was invited for re-examination in 2005 (sampling interval $K=10$), corresponding to $\approx 10\%$ of the original cohort. Other follow-up surveys were larger in scale, and efforts were made to include all participants originally enrolled in 1987. The corresponding response proportions were 77.7% ($n=3592$) in 1989, 84.8% ($n=3918$) in 1992, 82.1% ($n=3794$) in 1995, 65.3% ($n=3018$) in 2013, 60.1% ($n=2780$) in 2017, and 56.7% ($n=2621$) in 2023 (Figure S1). The primary reasons for loss to follow-up included pregnancy, military service, migration, death, and refusal to participate. Detailed information on the study design and protocol has been reported elsewhere.⁹ This study included participants who were examined at least 4 $\times$ (at least 2 $\times$ at ages 6–18 years and at least 2 $\times$ at ages ≥ 19 years) for BP values from childhood to mid-adulthood to accurately capture individual BP trajectories over time. As a result, 2508 participants were included for data analysis (Figure S1).

This study was conducted in accordance with the principles of the 1996 Declaration of Helsinki and received ethical approval from the Ethics Committee of the First Affiliated Hospital of Xi'an Jiaotong University (approval number: XJTU1AF2015LSL-047). It has been registered at ClinicalTrials.gov (URL: <https://www.clinicaltrials.gov>; unique identifier: NCT02734472). Written informed consent was obtained from all participants before each survey. For participants aged under 18 years, consent was provided by their parents/guardians.

BP Measurements

BP was measured following the same standardized protocol at each visit. Trained technicians measured the participants' BP using a standard mercury sphygmomanometer for the first 6 visits and a validated electronic sphygmomanometer (Omron M6, Kyoto, Japan)¹³ for the last 2 visits. When using the mercury sphygmomanometer, the first and fifth Korotkoff sounds were recorded as systolic BP (SBP) and diastolic BP (DBP), respectively. In participants aged under 12 years, the fourth Korotkoff sound was considered as DBP. After 3 to 5 minutes of rest, 3 consecutive BP readings were obtained on the right arm while seated. Cuff sizes were selected appropriately for the participant's upper arm circumference. The average of the 3 readings was calculated to define SBP and DBP for each examination.

Subclinical Target Organ Damage

Surrogate markers of subclinical target organ damage in mid-adulthood were assessed at the 2023 follow-up, including arterial stiffness (elevated brachial-ankle pulse wave velocity [baPWV]), LVH (elevated left ventricular mass index [LVMI]), and albuminuria (elevated urinary albumin-to-creatinine ratio).

Arterial stiffness was assessed using baPWV, a noninvasive and automated measure that has been widely used as a practical alternative to the gold-standard carotid-femoral PWV in large-scale epidemiological studies.^{2,14} The baPWV was measured using the volume-plethysmographic equipment BP-203RPE III (Nihon Colin, Tokyo, Japan). Participants were examined in a supine resting position, with ECG electrodes

placed on both wrists, cuffs wrapped on both upper arms and ankles, and a heart sound sensor positioned at the left sternal border in the fourth intercostal space. The mean baPWV of both sides was used for data analyses, and arterial stiffness was defined as $\text{baPWV} \geq 1400 \text{ cm/s}$.^{15,16}

Echocardiographic images were obtained by certified cardiologists using a Philips iE33 ultrasound system (Philips Ultrasound, Bothell, WA) equipped with an X3-1 transducer (1–3 MHz). Left ventricular structural parameters, including wall thickness and chamber dimensions, were subsequently measured in accordance with the American Society of Echocardiography guidelines.¹⁷ Left ventricular mass (LVM) was calculated with the formula: $\text{LVM} = 0.8 \times (1.04 \times [(\text{left ventricular internal diameter} + \text{interventricular septal thickness} + \text{left ventricular posterior wall thickness})^3 - \text{left ventricular internal diameter}^3]) + 0.6$.¹⁸ LVMI was calculated as LVM divided by 2.7 power of height.¹⁹ LVH was defined as $\text{LVMI} \geq 47 \text{ g/m}^{2.7}$ for women and $\geq 50 \text{ g/m}^{2.7}$ for men based on the 2024 European hypertension management guidelines.²⁰

A midstream morning urine sample was collected from each participant and stored at -80°C until use. Urinary albumin and creatinine levels were evaluated using a Hitachi 7600 auto-analyzer (Hitachi, Ltd, Tokyo, Japan). The urinary albumin-to-creatinine ratio (mg/g) was then calculated as urine albumin divided by urine creatinine. Albuminuria was defined as a urinary albumin-to-creatinine ratio $\geq 30 \text{ mg/g}$, following the Kidney Disease Outcomes Quality Initiative guidelines.²¹

Covariates

Covariates included age, sex (men versus women), body mass index, height, and resting heart rate at the baseline survey, and mid-adult body mass index, resting heart rate, occupation, marital status, educational attainment, physical activity, cigarette smoking, alcohol drinking, sleep quality, fasting blood glucose and lipid levels, and estimated glomerular filtration rate at the most recent survey. These covariates were selected based on their established or theoretical relevance to both BP exposure and target organ damage, rather than on data completeness.

Demographic and lifestyle factors were ascertained using standardized, self-administered questionnaires. Occupation was categorized as farmers or nonfarmers; marital status as married or divorced/unmarried/other; educational attainment as less than high school or high school and above; and physical activity as vigorous, moderate, light, or none. Cigarette smoking was defined as having smoked continuously or cumulatively for at least 6 months over the lifetime.²² Alcohol drinking was defined as daily consumption of alcohol (eg, beer, wine, or liquor) for at least 6 months.²² Sleep quality was captured using the Pittsburgh Sleep Quality Index questionnaire.²³

Body weight and height were measured in duplicate by trained staff, with participants wearing light clothing and no shoes, and the average values were used to calculate body mass index (kg/m^2). At the baseline examination (1987), resting heart rate was measured manually by pulse palpation, the standard method in large-scale epidemiological surveys at that time. At the 2023 follow-up, heart rate was measured 3 $\times$ using an automated sphygmomanometer (Omron M6), and the mean of the 3 readings was used. Venous blood samples were collected in the morning by experienced nurses after a minimum 8-hour fast. Fasting serum concentrations of glucose, total cholesterol, triglycerides, and creatinine were measured using

an autoanalyzer (Hitachi 7600, Hitachi, Tokyo, Japan). The estimated glomerular filtration rate was calculated using the following equation: estimated glomerular filtration rate (mL/min/1.73m²) = 186 × (serum creatinine)^{-1.154} × (age)^{-0.203} × 0.742 (if woman).

Statistical Analysis

Continuous variables were presented as median (interquartile range), and categorical variables as number (proportion). Group differences were evaluated using Wilcoxon rank-sum tests for continuous variables and χ^2 tests for categorical variables.

Group-based trajectory modeling, which has been widely used to identify distinct population-level BP trajectory classes (eg, 4 classes in our prior work^{9,24}), operates under the assumption that all individuals within a given class follow an identical underlying trajectory. This assumption limits the method's ability to generate individual-specific parameters and precludes direct estimation of individual BP levels and rates of change over time. In contrast, growth models (ie, multilevel or mixed-effects models for longitudinal data) incorporate both fixed and random effects, offering a more flexible framework for capturing individual trajectories. Fixed effects describe the mean BP trajectory for the overall sample, while random effects represent between-individual variability around this average. These models appropriately handle within-individual correlations inherent in repeated measurements and can accommodate missing or irregularly timed observations. We therefore used growth models to construct individual growth curves for SBP and DBP based on serial measurements obtained from childhood through mid-adulthood (ages 6–52 years).^{12,25} Separate models were fitted for men and women. Three potential polynomial models (linear, quadratic, and cubic) were sequentially fitted as a function of age. Each model is composed of both fixed and random effects of the corresponding terms. We refer to these modeled BP trajectories as growth curves in keeping with the standard terminology of growth models, which describes intraindividual change across time rather than biological growth per se.²⁶ Curve parameters were estimated using the maximum likelihood method. To reduce collinearity between linear and higher-order polynomial terms, age was centered at the cohort's mean age (25.7 years for men and 25.4 years for women). In addition, the squared age term was divided by 10 and the cubed age term by 20 to improve model fitting. Model selection was based on the Akaike information criterion and parameter significance (*P* values). Higher-order polynomial terms were excluded if they were not statistically significant, rendered lower-order terms nonsignificant, or did not reduce the Akaike information criterion value.^{12,25} Based on these criteria, cubic curves best fit the data, and the final model was described as follows:

$$BP_i = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i}) \text{age} + (\beta_2 + b_{2i}) \text{age}^2 + (\beta_3 + b_{3i}) \text{age}^3 + \varepsilon,$$

where β_0 , β_1 , β_2 , and β_3 represent the fixed effect parameters, b_{0i} , b_{1i} , b_{2i} , and b_{3i} are the individual-specific random effect parameters, and ε denotes the residual error. Sex differences in BP trajectories were assessed using a combined model that included interactions between sex and polynomial terms of age.

Model-estimated BP levels (mm Hg) and rates of change (instantaneous change, mm Hg/y) at each integer age were

derived from individual curve parameters and their first-order derivatives, respectively.^{12,25,27} Before logistic regression analyses, the model-estimated rates of change in SBP and DBP at each age point were adjusted for their corresponding levels through regression residual analyses to mitigate collinearity between the levels and rates of change in the same model. Then, both model-estimated levels and level-adjusted rates of change at each age point were standardized (mean=0; SD=1) separately for men and women to enable the comparison of effect sizes. Logistic regression analyses were used to estimate the associations of model-estimated BP levels and level-adjusted rates of change at each age point with subclinical target organ damage in mid-adulthood, adjusting for all covariates mentioned above. Odds ratios (ORs) and their corresponding 95% CIs per 1 SD increment in model-estimated BP levels and level-adjusted rates of change associated with subclinical target organ damage were calculated. The associations of model-estimated BP levels and level-adjusted rates of change with the outcomes were largely similar between men and women (data not shown). To ensure sufficient statistical power, we therefore conducted all logistic regression analyses in the pooled sample, with additional adjustment for sex.

To examine the robustness of our main findings, 6 sensitivity analyses were conducted: (1) excluding participants who reported ever using antihypertensive medications (*n*=132); (2) excluding all data from the 2005 examination, which involved a systematically sampled subset of 436 participants; (3) excluding participants with arterial stiffness (*n*=364) or albuminuria (*n*=175) identified in the 2017 survey (LVH was not assessed at that time); (4) excluding participants with a self-reported history of clinical cardiovascular or kidney events (including myocardial infarction, heart failure, stroke, peripheral artery disease, or chronic kidney disease; *n*=29) during follow-up, as documented in the 2023 questionnaire; (5) using multiple imputation for missing covariate data; and (6) utilizing restricted cubic splines with 4 knots instead of polynomial functions to model longitudinal BP patterns. For sensitivity analyses (1)–(4) and (6), individual growth curves were first reestimated following the specified exclusions or model specifications (data not shown). Then, we reexamined the associations of age-specific BP levels and rates of change with the outcomes.

To further validate the results, we calculated the area under the curve (AUC) for SBP and DBP across 2 exposure windows: adulthood-only, and the full life course (adulthood plus childhood and adolescence). The AUC was derived as the integral of each individual's fitted BP trajectory over the specified time period, which represents cumulative long-term BP burden. We then assessed whether inclusion of BP burden before age 18 years (AUC during childhood and adolescence) improved the prediction of mid-adult subclinical target organ damage beyond adult BP burden alone (AUC during adulthood), by comparing the predictive performance of life-course AUC (adulthood plus childhood and adolescence) with adulthood-only AUC using net reclassification improvement and integrated discrimination improvement analyses.

All data cleaning and analyses were performed using Stata, version 17.0 (StataCorp, College Station, TX) and R, version 4.4.1 (R Foundation for Statistical Computing, Vienna, Austria). Statistical significance was determined as a 2-tailed *P* < 0.05.

RESULTS

Characteristics of Study Participants

Among the 2508 participants included, 53.9% ($n=1352$) were men. The overall prevalence of arterial stiffness, LVH, and albuminuria in mid-adulthood was 25.8%, 6.5%, and 13.8%, respectively. As shown in the Table, men had significantly higher rates of arterial stiffness than women, whereas women exhibited significantly higher rates of LVH. In addition, from 1992 onward, men consistently had higher mean SBP and DBP levels than women. Significant sex differences were also observed in several characteristics in mid-adulthood, including cigarette smoking, alcohol drinking, physical activity, body mass index, resting heart rate, fasting levels of glucose and triglycerides, and estimated glomerular filtration rate (all $P<0.05$; Table). Comparisons of participant characteristics between those included and excluded are presented in Table S1.

Changes in BP Levels and Rates of Change Across the Life Course

As shown in Figure 1, SBP and DBP levels increased with age over the lifespan in both men and women. Significant sex differences were observed in the trajectories (P for interaction <0.001). The growth curves crossed at ≈ 12 years for SBP and 14 years for DBP, after which men consistently exhibited higher levels than women.

Tables S2 and S3 summarize detailed model-estimated SBP and DBP levels and rates of change at each age point from 6 to 52 years. SBP increased rapidly during childhood and adolescence, then increased slightly until around 40 years, and increased again thereafter. Similarly, the greatest DBP increase occurred in childhood and adolescence, with a further steady increase until 52 years. In addition, after ≈ 38 years for SBP and 42 years for DBP, rates of change among women surpassed those of men.

BP Levels and Rates of Change Across the Life Course and Subclinical Target Organ Damage

After adjusting for all potential covariates mentioned above, higher SBP levels at all ages were significantly associated with increased odds of arterial stiffness in mid-adulthood, with ORs (95% CIs) for each 1 SD increment in SBP levels increasing from 1.31 (1.15–1.50) at 6 years to 2.69 (2.31–3.13) at 52 years (Figure 2A). For LVH, significant associations were observed from age 8 (OR, 1.27 [95% CI, 1.00–1.60]) through 52 years (OR, 1.58 [95% CI, 1.27–1.97]; Figure 2B), and for albuminuria, ORs (95% CIs) were significant from 11 to 52 years, with values ranging from 1.19 (1.02–1.40) to 1.83 (1.58–2.13; Figure 2C).

As shown in Figure 2A, after further adjusting for SBP levels at each age point, higher level-adjusted rates

of change in SBP from 6 to 52 years were also significantly associated with increased risks of arterial stiffness in mid-adulthood. The ORs (95% CIs) per 1 SD increase in SBP change rates increased from 1.94 (1.69–2.24) at age 6 years to 2.11 (1.82–2.44) at age 13 years, and then declined to 1.13 (1.01–1.28) by age 52 years. For LVH, significant associations were observed between 6 and 26 years, with ORs per 1 SD increase in SBP change rates decreasing from 1.63 (1.29–2.05) to 1.28 (1.00–1.63; Figure 2B). Albuminuria exhibited a pattern similar to that of arterial stiffness, with ORs (95% CIs) per 1 SD increase in SBP change rates rising from 1.62 (1.38–1.89) at age 6 years to 1.71 (1.45–2.00) at age 14 years, and subsequently declining to 1.16 (1.01–1.35) by age 51 years (Figure 2C).

Notably, ORs (95% CIs) per 1 SD increase in SBP change rates in general exceeded those for corresponding SBP levels at each age point during childhood and adolescence (≈ 6 –25 years) for all 3 outcomes,²⁸ whereas there was an opposite trend in young adulthood (≈ 26 –39 years) or in middle age (40–52 years; Figure 2). The associations of model-estimated DBP levels and level-adjusted rates of change with arterial stiffness and albuminuria displayed nearly similar age-related change patterns to those observed for SBP (Figure S2). Although the associations between level-adjusted rates of change in DBP and LVH did not reach statistical significance after 13 years, the ORs remained above 1.00 until age 27 (Figure S2).

Sensitivity Analyses

Sensitivity analyses performed by excluding participants who reported a history of antihypertensive medication use (Figure S3), those who were a subpopulation for follow-up in 2005 (Figure S4), participants with preexisting arterial stiffness or albuminuria identified in 2017 (Figure S5), and individuals with clinical cardiovascular or kidney events in 2023 (Figure S6), as well as analyses using multiple imputation for missing values of covariates (Figure S7) and restricted cubic splines to model individual BP growth curves (Figure S8), yielded results largely consistent with the primary findings. In addition, net reclassification improvement and integrated discrimination improvement analyses revealed that BP burden accrued before 18 years of age (AUC across childhood and adolescence) provided incremental predictive value for mid-adult organ injury beyond BP burden in adulthood (AUC across adulthood; Tables S4 and S5), highlighting the important contribution of BP burden during childhood and adolescence to target organ damage in adulthood.

DISCUSSION

Leveraging 36 years of longitudinal data spanning from childhood to mid-adulthood, we first characterized

Table. Characteristics of Study Participants by Sex

Characteristic	Total (N=2508)	Men (n=1352)	Women (n=1156)
Childhood (first survey in 1987)			
Age, y	12.0 (9.5–14.0)	12.0 (9.0–14.0)	12.0 (10.0–14.0)
SBP, mm Hg	102.3 (96.0–110.0)	101.3 (95.3–110.0)	103.3 (97.0–110.6)
DBP, mm Hg	64.0 (59.3–70.7)	63.3 (59.3–70.7)	64.7 (60.0–70.7)
Resting heart rate, bpm	78.0 (72.0–84.0)	78.0 (72.0–84.0)	78.0 (72.0–84.0)
Height, cm	136.3 (124.0–147.6)	135.0 (123.0–147.0)	137.6 (124.9–148.4)
BMI, kg/m ²	15.9 (14.7–17.5)	15.8 (14.8–17.0)	16.0 (14.7–17.9)
Childhood to adulthood (1989–2017)			
SBP in 1989, mm Hg	106.7 (98.7–113.9)	106.0 (98.0–114.0)	107.3 (99.3–113.7)
DBP in 1989, mm Hg	67.3 (60.0–72.7)	66.7 (60.0–72.6)	68.0 (60.3–72.7)
SBP in 1992, mm Hg	112.0 (103.3–121.3)	114.0 (104.0–124.0)	111.3 (102.7–118.7)
DBP in 1992, mm Hg	69.3 (61.3–75.3)	69.3 (61.3–76.0)	69.3 (62.0–75.3)
SBP in 1995, mm Hg	117.3 (108.7–126.0)	120.7 (110.7–130.0)	114.0 (106.7–121.3)
DBP in 1995, mm Hg	73.3 (66.7–79.3)	74.7 (68.0–80.7)	72.0 (65.3–78.0)
SBP in 2005, mm Hg	121.2 (111.3–130.7)	126.7 (118.0–134.0)	113.3 (104.7–122.0)
DBP in 2005, mm Hg	78.7 (70.0–86.0)	82.0 (75.3–88.0)	73.3 (67.3–80.0)
SBP in 2013, mm Hg	120.0 (110.0–130.0)	124.0 (115.3–132.7)	113.3 (104.7–122.7)
DBP in 2013, mm Hg	79.3 (71.3–87.3)	83.0 (76.0–90.0)	74.7 (68.7–80.7)
SBP in 2017, mm Hg	121.0 (112.3–130.3)	124.7 (116.7–133.7)	115.7 (107.3–124.7)
DBP in 2017, mm Hg	75.7 (69.0–84.0)	79.0 (72.7–86.3)	71.7 (65.3–78.5)
Mid-adulthood (last survey in 2023)			
SBP, mm Hg	125.0 (114.7–136.7)	130.0 (120.0–140.3)	119.3 (110.0–130.3)
DBP, mm Hg	82.3 (75.0–90.7)	86.3 (79.3–94.0)	78.7 (72.7–86.0)
Resting heart rate, bpm	77.3 (70.0–85.3)	78.0 (70.3–86.3)	76.9 (70.0–84.0)
BMI, kg/m ²	24.5 (22.4–26.6)	25.3 (23.3–27.2)	23.6 (21.7–25.6)
Marital status, n (%)			
Married	1730 (69.0)	925 (68.4)	805 (69.6)
Divorced/unmarried or other	778 (31.0)	427 (31.6)	351 (30.4)
Educational attainment, n (%)*			
Less than high school	1197 (66.1)	623 (64.2)	574 (68.3)
High school and above	615 (33.9)	348 (35.8)	267 (31.7)
Occupation, n (%)*			
Farmers	331 (18.2)	174 (17.8)	157 (18.5)
Nonfarmers	1491 (81.8)	801 (82.2)	690 (81.5)
Cigarette smoking, n (%)*	754 (41.1)	723 (73.4)	31 (3.7)
Alcohol drinking, n (%)*	498 (27.2)	448 (45.6)	50 (5.9)
Physical activity, n (%)*			
Vigorous	190 (10.4)	157 (16.0)	33 (3.9)
Moderate	641 (35.1)	339 (34.6)	302 (35.6)
Light	754 (41.2)	348 (35.5)	406 (47.8)
None	244 (13.3)	136 (13.9)	108 (12.7)
Sleep quality score*	6.0 (3.0–8.0)	5.5 (3.0–8.0)	6.0 (3.0–8.0)
Fasting glucose, mmol/L*	5.3 (4.9–5.8)	5.4 (5.0–5.9)	5.2 (4.9–5.6)
Total cholesterol, mmol/L*	4.7 (4.2–5.3)	4.7 (4.2–5.3)	4.7 (4.2–5.3)
Triglycerides, mmol/L*	1.6 (1.1–2.3)	1.9 (1.3–2.6)	1.3 (1.0–1.8)
eGFR, mL/min per 1.73 m ² *	92.4 (84.3–100.7)	93.8 (85.8–102.6)	90.2 (82.8–98.8)
baPWV, cm/s*	1280.5 (1182.0–1406.0)	1307.5 (1202.5–1443.5)	1258.5 (1158.0–1373.5)

(Continued)

Table. Continued

Characteristic	Total (N=2508)	Men (n=1352)	Women (n=1156)
Arterial stiffness, n (%) [*]	414 (25.8)	255 (30.2)	159 (20.8)
LVMI, g/m ^{2.7*}	34.0 (28.7–40.2)	34.0 (28.7–39.9)	34.0 (28.6–40.7)
LVH, n (%) [*]	101 (6.5)	34 (4.2)	67 (9.1)
uACR, mg/g [*]	4.8 (1.8–14.5)	5.0 (1.8–15.5)	4.6 (1.7–14.2)
Albuminuria, n (%) [*]	223 (13.8)	118 (13.7)	105 (13.9)

Data are presented as median (interquartile range) for continuous variables and number (proportion) for categorical variables. baPWV indicates brachial-ankle pulse wave velocity; BMI, body mass index; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; LVH, left ventricular hypertrophy; LVMI, left ventricular mass index; SBP, systolic blood pressure; and uACR, urine albumin-to-creatinine ratio.
^{*}Some covariates and outcomes measured in adulthood had missing values.

the age-specific associations of both absolute BP levels and level-adjusted instantaneous rates of change (slopes) with mid-life arterial stiffness, LVH, and albuminuria. We found that the associations with absolute BP levels progressively strengthened with age, reaching their maximum in mid-adulthood. In contrast, the associations with level-adjusted rates of BP change accelerated during childhood, peaked in adolescence, and then declined thereafter. Notably, the associations between rates of change in BP and these outcomes were more pronounced compared with the corresponding BP levels during childhood and adolescence, whereas there was an opposite trend during young adulthood and middle age. Given that BP change rates reflect the tempo of cardiovascular risk accumulation, with steeper increases in early life predicting higher BP levels in adulthood, childhood and adolescence may represent crucial periods for optimizing BP trajectories to mitigate long-term cardiovascular and kidney risks.

Previous studies, including ours, have demonstrated that high BP levels in childhood and adolescence or certain life-course trajectories are significantly associated with increased risks of subclinical target organ damage later in life.^{6–9} Our current work extends these studies by examining the potentially important life-course periods of high BP exposure on subclinical target organ damage in mid-adulthood. We found that BP levels measured from childhood through adolescence were positively associated with subclinical target organ damage in mid-adulthood. The strength of these associations increased progressively over the follow-up period, peaking at the most recent examination. This pattern suggests that BP measurements obtained temporally closer to the outcome assessment may be more strongly associated with arterial stiffness, LVH, and albuminuria. Similarly, data from the Cardiovascular Risk in Young Finns Study, which assessed the relative contribution of BP levels at 3 life stages (childhood and adolescence, young adulthood, and mid-adulthood), found that BP levels at each life stage contributed to arterial stiffness later in life, with mid-adulthood BP showing the greatest contribution.⁶ However, limited evidence is available for comparison of BP levels at various ages with other subclinical outcomes such as LVH and albuminuria, as assessed in our study.

This observation is consistent with established BP tracking literature, which reports stronger tracking coefficients during adolescence compared with early childhood.²⁹ It is reasonable that BP levels in later life reflect a cumulative lifelong burden; therefore, their associations with adverse subclinical outcomes become stronger with increasing age. These findings provide further scientific rationale for maintaining normal BP levels from early life to achieve long-term cardiovascular and kidney health benefits.

Despite the moderate child-to-adult BP tracking, the rates of increase vary across life stages. Our data demonstrated that BP increased rapidly during childhood and adolescence, followed by a slower increase during adulthood, consistent with several previous studies.^{12,30} However, the clinical relevance of the rates of change in BP over the life course remains poorly understood. Some previous studies have explored the associations between the rates of increase in BP during certain life stages and subsequent cardiovascular outcomes by adjusting for BP levels.^{11,12,31} The British 1946 Birth Cohort study (N=1653) reported that the rapid rise of SBP in mid-adulthood (43–53 years) positively correlated with higher LVMI levels in later life.¹¹ The Bogalusa Heart Study (N=1154) identified age-dependent associations between rates of change in BP during childhood and adolescence and LVH in mid-adulthood, showing inverse associations for children aged 4 to 9 years (ORs, 0.75–0.82) and positive associations for adolescents aged 13 to 19 years (ORs, 1.29–1.46).¹² Moreover, findings from the Bogalusa Heart Study¹² suggested that childhood BP levels demonstrated a stronger association with adult outcomes than adolescent levels, a pattern that contrasts with other conventional BP tracking studies.²⁹ The authors pointed out that these inverse associations were unexpected and warranted further investigation. The studies above were performed in the United Kingdom or the United States, and they did not examine these associations across different life stages (eg, from childhood to adulthood). Our extended analyses in a separate Chinese sample demonstrated that rapid increases in BP at each age point (from childhood to mid-adulthood) were positively associated with mid-adult arterial stiffness and albuminuria, independent of BP levels at the corresponding age.

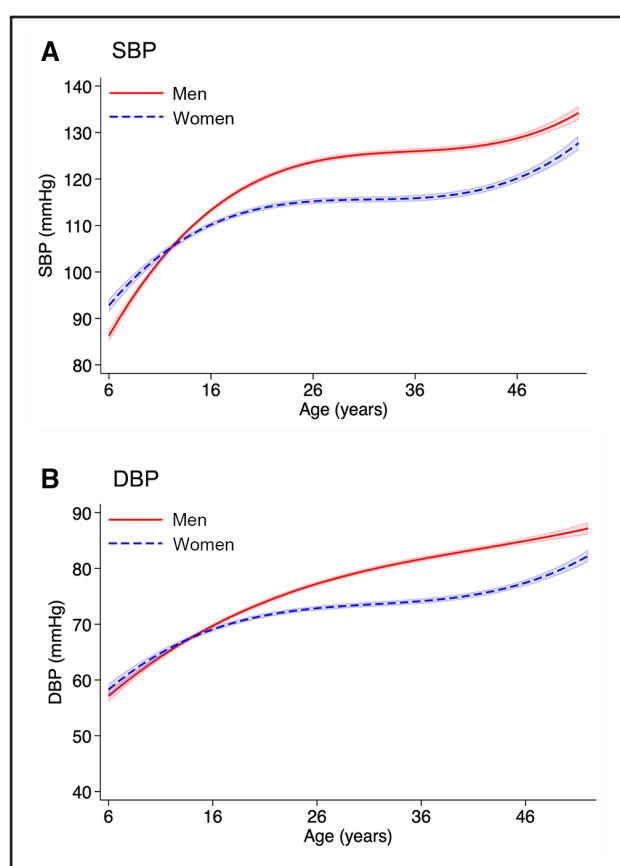


Figure 1. Sex-specific growth trajectories of blood pressure from childhood to mid-adulthood.

Trajectories are shown for systolic blood pressure (SBP; **A**) and diastolic blood pressure (DBP; **B**).

Crucially, it remains unclear whether specific temporal windows exist during which dynamic BP changes exert disproportionately detrimental effects on target organs. Our findings address this knowledge gap by elucidating a distinct bell-shaped susceptibility pattern: the magnitude of the associations between level-adjusted rates of BP change and subclinical outcomes intensified during childhood, peaked in adolescence, and attenuated thereafter. The observed stronger associations during childhood and adolescence (versus adulthood) are expected as the delta in BP becomes smaller with advancing age. In addition, the predictive value of BP change rates surpassed that of absolute BP levels during the first 2 decades of life. This phenomenon may be partially explained by the horse-racing hypothesis, which posits that an accelerated rate of increase drives subsequent higher absolute levels rather than the reverse.¹⁰ The peak vulnerability observed during adolescence likely reflects the synergistic impact of rapid somatic growth, pubertal hormonal fluctuations, and accelerated vascular remodeling characteristic of this developmental period,^{32,33} a phenomenon previously noted in the Bogalusa Heart Study.¹² Collectively, these findings suggest that childhood and adolescence represent crucial periods during

which accelerated BP increases program lasting structural and functional alterations in the cardiovascular system, thereby predisposing individuals to subclinical target organ damage that becomes more apparent in adulthood.

Of note, we found that both model-estimated BP levels and level-adjusted rates of change were more closely associated with adult arterial stiffness and albuminuria compared with LVH. This difference may arise from variations in the pathological response mechanisms between organs. Both arterial stiffness and albuminuria may directly reflect the impact of elevated BP on vascular function, often emerging in the earlier stages of CVD.^{4,34} Vascular dysfunction may subsequently elevate left ventricular afterload, which, over time, contributes to the development of LVH.³⁵ Indeed, only 6.5% of individuals in this study were diagnosed with LVH in mid-adulthood, lower than arterial stiffness (25.8%) and albuminuria (13.8%). Nonetheless, the trends in ORs for LVH were substantially consistent with those for arterial stiffness and albuminuria.

Strengths and Limitations

To our knowledge, this study is the first to examine when and how both BP levels and their dynamic changes across the life course exert the greatest impact on subsequent multiorgan subclinical damage (including vascular, cardiac, and renal markers), thereby yielding a more precise framework for prevention. However, this study has several limitations. First, the inclusion of only Chinese participants may restrict the generalizability of our findings to other racial/ethnic populations. In addition, observed differences in certain characteristics between included and excluded participants indicate potential selection bias. Second, the 2005 examination involved a systematic subsample ($\approx 10\%$ of the cohort), which could introduce bias. However, sensitivity analyses excluding the 2005 data did not alter our conclusions. Third, pharmacological treatments for hypertension may bias both the trajectory and association analyses. However, the exclusion of individuals with antihypertensive treatments at any follow-up showed essentially similar results to our main analyses. Fourth, the BP measurement device changed from a mercury sphygmomanometer to an electronic sphygmomanometer (Omron M6) in the last 2 visits. However, the Omron M6 has been validated against the mercury sphygmomanometer for BP measurement according to the European Society of Hypertension International Protocol.¹³ Fifth, the analysis was limited to outcome data collected at the final follow-up (2023), which precludes a direct evaluation of the temporal sequence between BP changes and the development of these outcomes. Nevertheless, after excluding participants identified with arterial stiffness or albuminuria in 2017 (LVH was not assessed at that time), our findings remained robust. Sixth, urine data were based on a morning spot

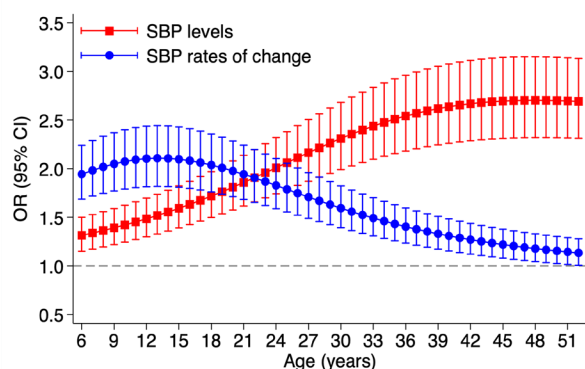
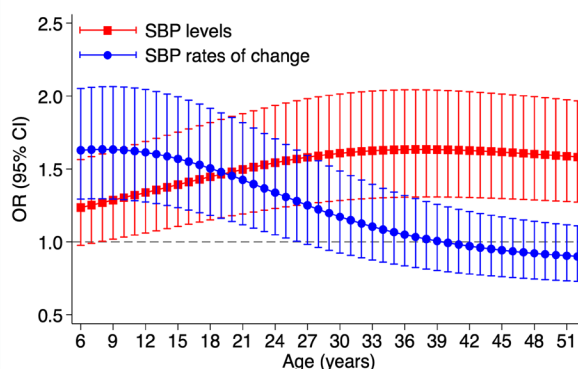
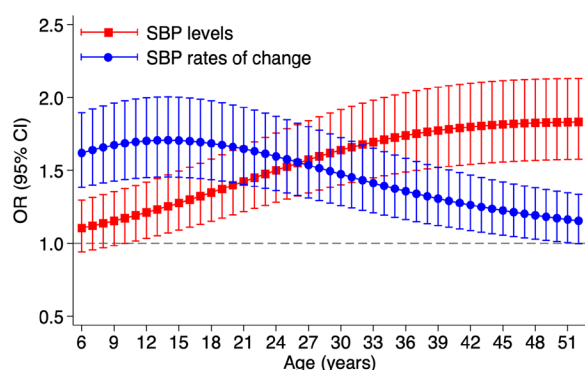
A Arterial stiffness**B** Left ventricular hypertrophy**C** Albuminuria

Figure 2. Age-specific associations of systolic blood pressure (SBP) levels and rates of change with mid-adult subclinical target organ damage.

Odds ratios (ORs) and 95% CIs are shown for arterial stiffness (**A**), left ventricular hypertrophy (**B**), and albuminuria (**C**). ORs correspond to per 1 SD increase in SBP levels or rates of change (slopes). Solid points represent the point estimates of ORs, and error bars denote the 95% CIs. The rates of change (slopes) represent the instantaneous change in BP at each age (mm Hg/y). All models were adjusted for baseline (age, sex, body mass index, height, and resting heart rate) and mid-adult covariates (body mass index, resting heart rate, occupation, marital status, educational attainment, physical activity, cigarette smoking, alcohol drinking, sleep quality, fasting serum levels of glucose and lipids, and estimated glomerular filtration rate). Models examining SBP change rates were additionally adjusted for concurrent SBP levels at each age point.

sample, which is not the gold standard to define albuminuria. Nevertheless, this approach has been validated to be associated with albumin excretion rate in a 24-hour urine collection and is considered an acceptable alternative for population-level screening of albuminuria.³⁶ Seventh, arterial stiffness was assessed using baPWV rather than the gold-standard carotid-femoral PWV. Nonetheless, previous studies have demonstrated a good correlation between baPWV and carotid-femoral PWV¹⁴ and have confirmed the predictive value of baPWV for cardiovascular events.² Eighth, this study was underpowered to directly assess hard clinical cardiovascular and kidney events due to the relatively small number of such cases in the cohort ($n=29$). Ninth, participants who died before the 2023 follow-up were excluded, which could introduce survivorship bias. However, among the 67 deaths with known causes, only 6 were attributable to CVD. Finally, as with any observational study, the possibilities

of reverse causation and residual confounding cannot be entirely excluded.

Our study has significant preventive and clinical implications. First, it provides compelling evidence supporting active BP management beginning in childhood and adolescence. Current strategies that primarily initiate hypertension prevention in adulthood may be inadequate to counter the cardiovascular risk burden that starts accumulating early in life. Notably, since absolute BP levels are inherently lower in youth than in adults over 30 years of age, the current diagnostic threshold ($\geq 130/80$ mm Hg) may fail to identify young individuals already at cardiovascular risk. This underscores the necessity for hypertension definitions to incorporate cumulative life-course risk rather than relying solely on cross-sectional, adult-derived thresholds. Second, our findings emphasize the critical need for dynamic surveillance of BP trajectories throughout childhood and adolescence. For children and

adolescents with predisposing risk factors, surveillance of their BP developmental patterns could facilitate the early identification of high-risk individuals and prompt timely interventions before subclinical target organ damage becomes established. Third, reliance on a single elevated BP reading may lead to misclassification of children whose long-term rate of BP change is at or below the population average. Some children may present with an initially high BP but subsequently follow a more benign trajectory, yielding a cumulative BP exposure lower than that suggested by the baseline value alone.

Perspectives

Our findings indicate that faster rates of BP increase during childhood and adolescence are associated with an increased risk of subclinical target organ damage in mid-adulthood, independent of corresponding BP levels. These findings highlight that childhood and adolescence are crucial periods for cardiovascular risk accrual. This suggests that preventive efforts should leverage this window by monitoring BP trajectories, not just single readings, to identify youth with accelerating BP for timely intervention, thereby laying a foundation for improved lifelong cardiorenal health.

ARTICLE INFORMATION

Received November 4, 2025; accepted April 15, 2026.

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Acknowledgments

The authors acknowledge all of the participants in our study and the staff responsible for conducting the Hanzhong Adolescent Hypertension Study.

Author Contributions

J. Mu and B. Xi designed and supervised this study. T. Guo performed the data analysis and prepared the tables and figures. L. Yang drafted the article. T. Guo, Y. Yang, J. Ren, M. He, C. Chu, Y. Yan, Y. Sun, D. Wang, G. Hu, M. Du, and H. Jia collected and interpreted the data. All authors critically revised the article for key intellectual content and approved the final version of the article. The corresponding authors attest that all the listed authors meet the authorship criteria and that no others meeting the criteria have been omitted.

Sources of Funding

This work was supported by the National Natural Science Foundation of China (82070437, 82173538, and 82473653), the Key Research and Development Program of Shaanxi (2023-ZDLSF-50), the Clinical Research Award of the

First Affiliated Hospital of Xi'an Jiaotong University (XJTU1AF-CRF-2023-004, XJTU1AF-CRF-2024-003, XJTU1AF-CRF-2024-007), the International Joint Research Center for Cardiovascular Precision Medicine of Shaanxi Province (2020GHJD-14), the Distinguished Professor of Changjiang Scholar Awards Program (T2023116), and the Natural Science Foundation of Shandong Province (ZR2024QH194). The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the article; and decision to submit the article for publication.

Disclosures

None.

Supplemental Material

Tables S1–S5

Figures S1–S8

Major Resources Table

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